1. (3 points) This exercise is a training in literature search. Find the following e-book in the TUB online library: Folkmar Bornemann, *Numerical Linear Algebra. A Concise Introduction with MATLAB and Julia*, Springer Verlag, 2018. Answer the following questions, using this book:

   (a) What is the ISBN and the DOI of the book?

Title	Numerical Linear Algebra: A Concise Introduction with MATLAB and Julia
Author	Bornemann, Folkmar
Subjects	Algebra & number theory Linear and Multilinear Algebras, Matrix Theory Mathematics Mathematics and Statistics Numerical Analysis
Description	This book offers an introduction to the algorithmic-numerical thinking using basic problems of linear algebra. By focusing on linear algebra, it ensures a stronger thematic coherence than is otherwise found in introductory lectures on numerics. The book highlights the usefulness of matrix partitioning compared to a component view, leading not only to a clearer notation and shorter algorithms, but also to significant runtime gains in modern computer architectures. The algorithms and accompanying numerical examples are given in the programming environment MATLAB, and additionally - in an appendix - in the future-oriented, freely accessible programming language Julia. This book is suitable for a two-hour lecture on numerical linear algebra from the second semester of a bachelor's degree in mathematics.
Related Titles	Springer Undergraduate Mathematics Series
Publisher	Cham: Springer Nature
Publication Date	2018
Edition	1st ed. 2018.
Format	157
Language	English
Identifier	ISSN: 1615-2085 ISBN: 9783319742229 ISBN: 3319742221 ISBN: 3319742213 ISBN: 9783319742212 EISSN: 2197-4144 EISBN: 9783319742229 EISBN: 3319742221 DOI: 10.1007/978-3-319-74222-9

   (b) What is, according to Bornemann, the "Zeroth Law of Numerical Analysis"?

   If theoretical analysis struggles for $a=0$, numerical analysis does so for $a \approx 0$

   (c) Which "professional blunder" should not be attempted according to the LAPACK manual?

   According to LAPACK manual,
   one should not attempt to solve a system of eq. $Ax=b$ by first computing A^{-1} and then forming the matrix-vec. product $x=A^{-1}b$

2. (2 points) This exercise is a training in literature search.

   (a) Which book of A. S. Householder is cited in J. L. Rigal and J. Gaches, *On the Compatibility of a Given Solution With the Data of a Linear System*, J. ACM, 3 (1967), pp. 543–548? (State the complete reference.)

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Received July, 1966

(b) Which article of L. Collatz is cited in F. L. Bauer and A. S. Householder, *Moments and characteristic roots*, Numerische Mathematik, 2 (1960), pp. 42–53? (State the complete reference.)

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3. (9 points) Prove the following assertions, using mathematical induction:

(a) If n is a positive integer and $r \in \mathbb{R} \setminus \{1\}$, then

$$\sum_{k=0}^{n-1} r^k = \frac{r^n - 1}{r - 1}.$$

[Base-Case],

$$n=1, \quad \underbrace{\sum_{k=0}^{0} r^k = r^0 = 1}_{\text{left side}} = \underbrace{\frac{r^0-1}{r-1}}_{\text{right side}} = 1$$

∴ formula holds for $n=1$

[Inductive step]

$$\sum_{k=0}^{n-1} r^k = \frac{r^n-1}{r-1}$$

Consider $n+1$

$$\sum_{k=0}^{0} r^k = \left(\sum_{k=0}^{n-1} r^k\right) + r^n = \frac{r^n-1}{r-1} + r^n$$

$$\frac{r^n-1}{r-1} + r^n = \frac{r^n-1+r^n(r-1)}{(r-1)}$$

$$= \frac{\cancel{r^n} - 1 + r^{n+1} - \cancel{r^n}}{r-1}$$

$$= \frac{r^{n+1}-1}{r-1}$$

∴ formula holds for $n+1$

By induction, the identity is true for every positive integer n.

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Date

Exam

Subject

Importance

Review

(b) Every integer $n \geq 4$ satisfies $n! > 2^n$.

Base case: $n=4$,

$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$ and $2^4 = 16$

$\because 24 \geq 16$, statement holds for $n=4$

Inductive step

Suppose for some int. $n \geq 4$ that satisfies $n! > 2^n$

Consider $n+1$

$(n+1)! = (n+1)(n)! > (n+1) \cdot 2^n$

Since $n \geq 4$, $n+1 > 2$

$(n+1) 2^n > 2 \cdot 2^n = 2^{n+1}$

$(n+1)! > 2^{n+1}$

$\therefore$ Property holds for $n+1$

By induction the inequality $n! > 2^n$ holds for all int $n \geq 4$

(c) Prove that, for the Fibonacci sequence defined by $F_1 = 1$, $F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n > 2$, the following holds for every integer $n \geq 1$:

$$\sum_{k=1}^{n} F_k = F_{n+2} - 1.$$

Base case: $n=1$

$\underbrace{\sum_{k=1}^{1} F_k = F_1 = 1}_{\text{right}} = \underbrace{F_3 - 1 = F_2 + F_1 - 1 = 1}_{\text{left}}$

Inductive step

Assume some $n \geq 1$

$\sum_{k=1}^{n} F_k = F_{n+2} - 1$

Then for $n+1$,

$\sum_{k=1}^{n+1} F_k = \left(\sum_{k=1}^{n} F_k\right) + (F_{n+1}) = (F_{n+2} - 1) + (F_{n+1})$

$= F_{n+1} + F_{n+2} - 1$

By Fibonacci recur. $F_{n+3} = F_{n+1} + F_{n+2}$

$\therefore \sum_{k=1}^{n+1} F_k = F_{n+3} - 1$

$\rightarrow$ holds for $n+1$. $\therefore$ By induction, it holds for every int. $n \geq 1$